Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

2022

Bachelor in Computer Applications

Course Title: **Computer Networking**

Code No: **CACS303** Semester: **V**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

***Note:** Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

2. What is digital signature and how it works?

3. Why do we need DNS when we can directly use an IP address? Explain the process to resolve the domain www.tribhuvan-university.edu.np?

4. What do you mean by IP datagram fragmentation?

5. Explain the mechanism of bit and byte stuffing with suitable example.

6. Explain the causes of transmission impairment in data communication.

7. What is flow control? Explain Go-Back-N ARQ with suitable diagram.

8. Explain in detail about the structure of IPv6 address. Differentiate between IPv4 and IPv6 address.

Group C

Attempt any TWO questions[2x10=20]

9. Explain how slotted Aloha improves the performance of system over pure Aloha. A bit stream of 100101010 is transmitted using CRC method.The generator polynomial is $x^3 + 1$. Show the actual bit string transmitted and if fourth bit from LSB is altered, show the error detection in receiver side.

10. Why do we need layered protocol architecture? Discuss each layer of TCP/IP architecture along with function of each layer. Write the protocols used in each layer of TCP/IP.

11. Why do we need subnetting? What is the new subnet mask? Write subnet ID and broadcast address of each subnet if you divide a class C network (192.6.3.0–192.6.3.255) into 4 different subnets?